Calvin Hou

Seattle, WA | 206-601-6421 | chou61@gatech.edu | calvinhou.com (Portfolio) | linkedin.com/in/houcalvin

Education

B.S. in Electrical Engineering, GPA: 3.9 — May 2028 (Expected)
Georgia Institute of Technology — Atlanta, GA

Technical Skills

Embedded Systems: Schematic Design & PCB Layout (Altium, KiCad), Power Electronics, Soldering, Analog Circuit Design, Digital Circuit Design, Battery Management Systems, Signal Integrity, Board-Level Debug, STM32, CAN FD, UART, SPI, I2C
Tools & Languages: C, C++, Embedded C, Python, SystemVerilog, VHDL, Git, CAD, LTspice, Mandarin Chinese
Test & Lab Equipment: Oscilloscope, Logic Analyzer, Multimeter, DAQ Systems, Soldering, Wire Harnessing & Crimping
Coursework: Digital System Design + Lab, Prog. for Hardware/Software Systems, Signal Processing, Circuits & Electronics

Experience

Undergraduate Research Assistant — Aug. 2026 – Present
Yeo Lab, Georgia Institute of Technology — Atlanta, GA

- Designed a 2-layer rigid-flex PCB for wearable core body temperature estimation research, integrating ECG, photoplethysmography (PPG), and temperature sensing with BLE connectivity; sized flex dielectric thickness to achieve a 50 Ω controlled-impedance RF trace from the MCU to the antenna
- Leading end-to-end development of a **64-channel** EMG acquisition PCB in KiCad, from schematic capture and layout through bring-up, generating data to train a neural network that reconstructs muscle activity using fewer channels

Firmware Research and Design Intern — Jun. 2026 – Aug. 2026
Delta Electronics — Bangkok, Thailand

- Diagnosed EMI-induced signal corruption causing **critical current sensing inconsistencies** in data center PSUs; validated **21** IIR and glitch filter configurations in STM32 embedded C to eliminate erroneous fault readings
- Identified siloed MCAL implementations across Delta's PSU lines and built a standardized library of **6** modules (PWM, ADC, DAC, CAN FD, UART, Watchdog), eliminating redundant firmware rewrites for **8** engineers across **6** product lines
- Led an intern subteam to build Python hardware-in-the-loop test suites across the **6** MCAL modules, validating **3,000+** configurations and generating **UL 60730-1** certification evidence for regulatory submission

Hardware Research and Design Intern — Aug. 2024 – Sep. 2024
Asia Vital Components — New Taipei City, Taiwan

- Integrated thermistors and DAQ systems into liquid cooling hardware, characterized thermal performance under 1kW dummy heat loads and surfaced airflow and enclosure sealing gaps that informed R&D design iterations
- Assembled and pressure-tested coolant distribution units and coolant pumps for a rack-mounted sidecar cooling system supporting **NVIDIA Blackwell** GPU infrastructure in AI data centers

Projects

Avionics Responsible Engineer — Jan. 2026 – Present
Yellow Jacket Space Program (YJSP) — Atlanta, GA

- Developing a 240V HV BMS as Responsible Engineer; down-selected the cell monitoring IC across key parameters and designing a validation test board ahead of full schematic and PCB layout
- Led design of the STM32-based control PCB for the Control Test Vehicle from schematic to fab in Altium, integrating a multi-rail power system (22.2V → 12V/4.4V/3.3V) with per-rail ADC telemetry, an ADIN1200 Ethernet PHY hardened for **high-EMI** signal integrity, and drivers for **2x** linear actuators, **4x** RCS fans, and **2x** brushless ESCs

Electrical Systems Member — Aug. 2025 – Present
Hytech Racing (Electric Formula at Georgia Tech) — Atlanta, GA

- Led integration of **6** DS18B20 sensors across battery pack modules (schematic, 1-Wire firmware, on-vehicle validation), achieving **FSAE EM.3.3** compliance
- Introduced a J1772 inlet to the accumulator charging system, enabling **240V** access at competitions and cutting charge time by **50%**

1S USB-C PD 3.1 Battery Charger | *Altium, ESP32, Embedded C* — Jan. 2026 – August 2026

- Designed a **4**-channel, **15W** 1S Li-ion charger PCB in Altium with thermal and EMI mitigation; wrote ESP32 firmware for an OLED display delivering real-time SOC and thermal monitoring

Honors & Awards

- **Amazon Future Engineer Externship**, **1st Place, Technical Skills Track**; selected from **3,000+** Seattle-area applicants
- **Faculty Honors & Dean's List**, Georgia Institute of Technology
- **FIRST Robotics Competition** | Team 9442, Miso Mechanics (Founder, 2023)
  - **District Event Winner**; founded the team from scratch, raising **$45,000+** in sponsorships; ranked **top 4%** globally; fielded **4 robots** with **zero electrical failures**
  - **6 awards**: **2x** Rookie Inspiration, Excellence in Engineering, Industrial Design, Rising All Star, Rookie All Star